CSE370 Lab Assignment - 1

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Section: 04

MySQL commands;

- mysql -u root -p
- “”
- show databases;
- create database **CSE370_lab_assignment_1**;
- use **CSE370_lab_assignment_1**;

Task-1:

- create table **Developers**(

  member_id int,

  name varchar(30),

  email varchar(60),

  influence_count int,

  joining_date date,

multiplier int);

- show tables;
- describe **Developers**;

```
Setting environment for using XAMPP for Windows.
Hp@LAPTOP-HJV0I7NE c:\xampp
#  mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 11
Server version: 10.4.28-MariaDB mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> create database CSE370_lab_assignment_1;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> use CSE370_lab_assignment_1;
Database changed
MariaDB [CSE370_lab_assignment_1]> create table Developers(
    -> member_id int,
    -> name varchar(30),
    -> email varchar(60),
    -> influence_count char(6),
    -> joining_date date,
    -> multiplier int);
Query OK, 0 rows affected (0.228 sec)

MariaDB [CSE370_lab_assignment_1]> show tables;
+-----------------------------------+
| Tables_in_cse370_lab_assignment_1 |
+-----------------------------------+
| developers                        |
+-----------------------------------+
1 row in set (0.001 sec)

MariaDB [CSE370_lab_assignment_1]> describe Developers;
+-----------------+-------------+------+-----+---------+-------+
| Field           | Type        | Null | Key | Default | Extra |
+-----------------+-------------+------+-----+---------+-------+
| member_id       | int(11)     | YES  |     | NULL    |       |
| name            | varchar(30) | YES  |     | NULL    |       |
| email           | varchar(60) | YES  |     | NULL    |       |
| influence_count | char(6)     | YES  |     | NULL    |       |
| joining_date    | date        | YES  |     | NULL    |       |
| multiplier      | int(11)     | YES  |     | NULL    |       |
+-----------------+-------------+------+-----+---------+-------+
6 rows in set (0.655 sec)
```

- insert into **Developers** values

  (1,'Taylor Otwell','otwell@laravel.com',739360,'2020-6-10',10),

  (2,'Ryan Dahl','ryan@nodejs.org',633632,'2020-04-22',10),

  (3,'Brendan Eich','eich@javascript.com',939570,'2020-05-07',8),

  (5,'Evan You','you@vuejs.org',982630,'2020-06-11',7),

  (6,'Rasmus Lerdorf','lerdorf@php.net',937927,'2020-06-3',8),

  (7,'Guido van Rossum','guido@python.org',968827,'2020-07-18',19),

  (8,'Adrian Holovaty','adrian@djangoproject.com',570724,'2020-05-07',5),

  (9,'Simon Willison','simon@djangoproject.com',864615,'2020-04-30',4),

  (10,'James Gosling','james@java.com',719491,'2020-05-18',5),

  (11,'Rod Johnson','rod@spring.io',601744,'2020-05-18',7),

  (12,'Satoshi Nakamoto','nakamoto@blockchain.com',630488,'2020-05-10',10);

```
MariaDB [CSE370_lab_assignment_1]> insert into Developers values
    -> (1,'Taylor Otwell','otwell@laravel.com',739360,'2020-6-10',10),
    -> (2,'Ryan Dahl','ryan@nodejs.org',633632,'2020-04-22',10),
    -> (3,'Brendan Eich','eich@javascript.com',939570,'2020-05-07',8),
    -> (5,'Evan You','you@vuejs.org',982630,'2020-06-11',7),
    -> (6,'Rasmus Lerdorf','lerdorf@php.net',937927,'2020-06-3',8),
    -> (7,'Guido van Rossum','guido@python.org',968827,'2020-07-18',19),
    -> (8,'Adrian Holovaty','adrian@djangoproject.com',570724,'2020-05-07',5),
    -> (9,'Simon Willison','simon@djangoproject.com',864615,'2020-04-30',4),
    -> (10,'James Gosling','james@java.com',719491,'2020-05-18',5),
    -> (11,'Rod Johnson','rod@spring.io',601744,'2020-05-18',7),
    -> (12,'Satoshi Nakamoto','nakamoto@blockchain.com',630488,'2020-05-10',10);
Query OK, 11 rows affected (5.896 sec)
Records: 11  Duplicates: 0  Warnings: 0

MariaDB [CSE370_lab_assignment_1]> Select*from Developers;
+-----------+-----------------+--------------------------+-----------------+--------------+------------+
| member_id | name            | email                    | influence_count | joining_date | multiplier |
+-----------+-----------------+--------------------------+-----------------+--------------+------------+
|         1 | Taylor Otwell   | otwell@laravel.com       | 739360          | 2020-06-10   |         10 |
|         2 | Ryan Dahl       | ryan@nodejs.org          | 633632          | 2020-04-22   |         10 |
|         3 | Brendan Eich    | eich@javascript.com      | 939570          | 2020-05-07   |          8 |
|         5 | Evan You        | you@vuejs.org            | 982630          | 2020-06-11   |          7 |
|         6 | Rasmus Lerdorf  | lerdorf@php.net          | 937927          | 2020-06-03   |          8 |
|         7 | Guido van Rossum | guido@python.org         | 968827          | 2020-07-18   |         19 |
|         8 | Adrian Holovaty | adrian@djangoproject.com | 570724          | 2020-05-07   |          5 |
|         9 | Simon Willison  | simon@djangoproject.com  | 864615          | 2020-04-30   |          4 |
|        10 | James Gosling   | james@java.com           | 719491          | 2020-05-18   |          5 |
|        11 | Rod Johnson     | rod@spring.io            | 601744          | 2020-05-18   |          7 |
|        12 | Satoshi Nakamoto | nakamoto@blockchain.com  | 630488          | 2020-05-10   |         10 |
+-----------+-----------------+--------------------------+-----------------+--------------+------------+
11 rows in set (0.000 sec)

MariaDB [CSE370_lab_assignment_1]>
```

Task-2:

- Alter table **Developers**

  change column **influence_count followers** int;

```
MariaDB [CSE370_lab_assignment_1]> Alter table Developers
    -> change column influence_count followers int;
Query OK, 11 rows affected (1.213 sec)
Records: 11  Duplicates: 0  Warnings: 0

MariaDB [CSE370_lab_assignment_1]> Select*from Developers;
+-----------+------------------+--------------------------+-----------+--------------+------------+
| member_id | name             | email                    | followers | joining_date | multiplier |
+-----------+------------------+--------------------------+-----------+--------------+------------+
|         1 | Taylor Otwell    | otwell@laravel.com       |    739360 | 2020-06-10   |         10 |
|         2 | Ryan Dahl        | ryan@nodejs.org          |    633632 | 2020-04-22   |         10 |
|         3 | Brendan Eich     | eich@javascript.com      |    939570 | 2020-05-07   |          8 |
|         5 | Evan You         | you@vuejs.org            |    982630 | 2020-06-11   |          7 |
|         6 | Rasmus Lerdorf   | lerdorf@php.net          |    937927 | 2020-06-03   |          8 |
|         7 | Guido van Rossum | guido@python.org         |    968827 | 2020-07-18   |         19 |
|         8 | Adrian Holovaty  | adrian@djangoproject.com |    570724 | 2020-05-07   |          5 |
|         9 | Simon Willison   | simon@djangoproject.com  |    864615 | 2020-04-30   |          4 |
|        10 | James Gosling    | james@java.com           |    719491 | 2020-05-18   |          5 |
|        11 | Rod Johnson      | rod@spring.io            |    601744 | 2020-05-18   |          7 |
|        12 | Satoshi Nakamoto | nakamoto@blockchain.com  |    630488 | 2020-05-10   |         10 |
+-----------+------------------+--------------------------+-----------+--------------+------------+
11 rows in set (0.002 sec)

MariaDB [CSE370_lab_assignment_1]>
```

Task-3:

- Update **Developers** set **followers** = **followers** + 10;

  Select*from **Developers**;

```
MariaDB [CSE370_lab_assignment_1]> Update Developers set followers = followers + 10;
Query OK, 11 rows affected (0.169 sec)
Rows matched: 11  Changed: 11  Warnings: 0

MariaDB [CSE370_lab_assignment_1]> Select*from Developers;
+-----------+------------------+--------------------------+-----------+--------------+------------+
| member_id | name             | email                    | followers | joining_date | multiplier |
+-----------+------------------+--------------------------+-----------+--------------+------------+
|         1 | Taylor Otwell    | otwell@laravel.com       |    739370 | 2020-06-10   |         10 |
|         2 | Ryan Dahl        | ryan@nodejs.org          |    633642 | 2020-04-22   |         10 |
|         3 | Brendan Eich     | eich@javascript.com      |    939580 | 2020-05-07   |          8 |
|         5 | Evan You         | you@vuejs.org            |    982640 | 2020-06-11   |          7 |
|         6 | Rasmus Lerdorf   | lerdorf@php.net          |    937937 | 2020-06-03   |          8 |
|         7 | Guido van Rossum | guido@python.org         |    968837 | 2020-07-18   |         19 |
|         8 | Adrian Holovaty  | adrian@djangoproject.com |    570734 | 2020-05-07   |          5 |
|         9 | Simon Willison   | simon@djangoproject.com  |    864625 | 2020-04-30   |          4 |
|        10 | James Gosling    | james@java.com           |    719501 | 2020-05-18   |          5 |
|        11 | Rod Johnson      | rod@spring.io            |    601754 | 2020-05-18   |          7 |
|        12 | Satoshi Nakamoto | nakamoto@blockchain.com  |    630498 | 2020-05-10   |         10 |
+-----------+------------------+--------------------------+-----------+--------------+------------+
11 rows in set (0.001 sec)
```

Task-4:

- Select **name,((followers*100/1000000) *(multiplier*100/20))/100** as **Efficiency**

  from **Developers**;

```
MariaDB [CSE370_lab_assignment_1]>  Select name,((followers*100/1000000) * (multiplier*100/20))/100 as Efficiency
    -> from Developers;
+------------------+-----------------+
| name             | Efficiency      |
+------------------+-----------------+
| Taylor Otwell    | 36.968500000000 |
| Ryan Dahl        | 31.682100000000 |
| Brendan Eich     | 37.583200000000 |
| Evan You         | 34.392400000000 |
| Rasmus Lerdorf   | 37.517480000000 |
| Guido van Rossum | 92.039515000000 |
| Adrian Holovaty  | 14.268350000000 |
| Simon Willison   | 17.292500000000 |
| James Gosling    | 17.987525000000 |
| Rod Johnson      | 21.061390000000 |
| Satoshi Nakamoto | 31.524900000000 |
+------------------+-----------------+
11 rows in set (0.044 sec)

MariaDB [CSE370_lab_assignment_1]>
```

Task-5:

- select **name, email** from **Developers**

  order by **followers** desc limit 5;

```
MariaDB [CSE370_lab_assignment_1]> select name, email from Developers
    -> order by followers desc limit 5;
+------------------+-------------------------+
| name             | email                   |
+------------------+-------------------------+
| Evan You         | you@vuejs.org           |
| Guido van Rossum | guido@python.org        |
| Brendan Eich     | eich@javascript.com     |
| Rasmus Lerdorf   | lerdorf@php.net         |
| Simon Willison   | simon@djangoproject.com |
+------------------+-------------------------+
5 rows in set (0.001 sec)
```

Task-6:

- Select **name** from **Developers** where **followers** < 700000

  and **multiplier** = (Select max(**multiplier**) from **Developers** where **followers** < 700000);

```
MariaDB [CSE370_lab_assignment_1]> Select name from Developers where followers < 700000
    -> and multiplier = (Select max(multiplier) from Developers where followers < 700000);
+------------------+
| name             |
+------------------+
| Ryan Dahl        |
| Satoshi Nakamoto |
+------------------+
2 rows in set (0.001 sec)
```

Task-7:

- Select avg(**followers**) from **Developers**

  where **joining_date** < '2020-06-11';

```
MariaDB [CSE370_lab_assignment_1]> Select avg(followers) from Developers
    -> where joining_date < '2020-06-11';
+----------------+
| avg(followers) |
+----------------+
|    737515.6667 |
+----------------+
1 row in set (0.001 sec)
```

Task-8:

- Select **member_id**, **name**, **email**, **followers** from **Developers**

  where **email** like '%.com' or **email** like '%.net';

```
MariaDB [CSE370_lab_assignment_1]> Select member_id, name, email, followers from Developers
    -> where email like '%.com' or email like '%.net';
+-----------+------------------+-------------------------+-----------+
| member_id | name             | email                   | followers |
+-----------+------------------+-------------------------+-----------+
|         1 | Taylor Otwell    | otwell@laravel.com      |    739370 |
|         3 | Brendan Eich     | eich@javascript.com     |    939580 |
|         6 | Rasmus Lerdorf   | lerdorf@php.net         |    937937 |
|         8 | Adrian Holovaty  | adrian@djangoproject.com |    570734 |
|         9 | Simon Willison   | simon@djangoproject.com |    864625 |
|        10 | James Gosling    | james@java.com          |    719501 |
|        12 | Satoshi Nakamoto | nakamoto@blockchain.com |    630498 |
+-----------+------------------+-------------------------+-----------+
7 rows in set (0.071 sec)
```